From a research question to a context atlas

a Define the research question

Existing evidence
Findings from prior studies

Researcher question
An expected X–Y relationship

Study specification

X	Focal predictor	**Y**	Outcome of interest
X contrast	Lower → Higher	**Keep fixed**	Unit, time and core facts

Search objective
Find conditions that strengthen, weaken or reverse the relationship
5 rounds · 2 selected pairs per round

Search specification

b Search, assess and extend the atlas

1 Propose and select

Propose four
Observable conditions
Mechanism and rival account
Possible empirical test

Review designs
Feasibility and consistency
Distinctiveness

Select two
Relationship-change potential
Attribute and concept coverage

Two selected pairs per round
Ten comparisons per study

2 Compare matched settings

Same X contrast, Y definition, unit and time horizon
Only the highlighted contextual condition differs

Reference setting

Approval	Central executive
Information	Same access
Resources	Same constraints

Same lower-to-higher X contrast

Changed setting

Approval	Local consent
Information	Same access
Resources	Same constraints

Same lower-to-higher X contrast

Task and facts only · no search history

Assessment 1
Reference → Changed
Both settings receive an X–Y rating
−100 0 +100

Assessment 2
Changed → Reference
Both settings receive an X–Y rating
−100 0 +100

The second assessment also reverses fact order

Difference in X–Y ratings: changed − reference

3 Add contextual comparisons

A growing context atlas
Illustrative contextual comparisons
Each node records one matched comparison

Approval
Central executive
→ Local consent

Information
Shared access
→ Uneven access

Time pressure
Low urgency
→ High urgency

Commitment
Reversible
→ Fixed

Resource slack
Limited
→ Ample

Visibility
Private
→ Public

Assessed Proposed
Arrows show search progression

c Carry research questions forward

1 Keep a private research memory

Available to the planner

- **Proposed designs** — Conditions and mechanisms
- **Selected pairs** — Reviewed comparisons
- **Numerical results** — Ratings and contextual changes

Evaluators receive no private memory

2 Plan the next round

Use what earlier comparisons reveal
Pursue unresolved mechanisms
Avoid repeated comparisons
Explore promising attributes

Workflow Memory feedback

3 Develop empirical studies

Candidate studies
Clear contextual contrasts
Measurable conditions and testable claims

Empirical testing is the next stage